

ABHIJEET

PhD, Aerospace Engineering, Texas A&M University

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RESEARCH INTERESTS

- Learning Based Control
- Imitation Learning
- Motion Planning
- Reinforcement Learning
- Optimal Control Theory
- ILQR/DDP
- Numerical Optimization
- Model Predictive Control
- Feedback Control System

EXPERIENCE

Learning based Control

Texas A&M University

June 2025 - Ongoing College Station, TX

- Applied **Control Barrier Functions** (CBFs) to enforce constraints in **Reinforcement learning policies: SAC, TD3, and PPO** for robotic tasks.
- Working on developing **imitation learning** pipelines for various robotic tasks.
- Working on making imitation learning robust by introducing **LQR feedback-control** near the expert trajectory.

Graduate Research Assistant

Texas A&M University

August 2023 – Ongoing College Station, TX

- Developed **iterative perturbation feedback solvers** to solve optimal control problems with closed-form solutions.
- Implemented nonlinear programming solvers (**Sequential Quadratic Programming/Interior Point methods**) and benchmarked their performance against perturbation feedback technique (**iLQR**), reducing time complexity from $\mathcal{O}(n^3)$ to $\mathcal{O}(n)$.
- Applied **perturbation feedback methods** to solve optimal control problems with improved stability of the solution over existing methods.
- Formulated an innovative approach to solve **infinite horizon optimal control problems** using perturbation feedback, demonstrating superior computational efficiency and convergence over sliding horizon **Model Predictive Control**.
- Devised novel perturbation feedback control algorithms to solve complex **astrodynamics** problems, yielding enhanced precision in **spacecraft attitude** and **orbit control** under disturbances.
- Solved **constrained** optimal control problems for space applications using perturbation feedback.
- Developed new algorithm to solve optimal control problem with constraints using iLQR."Link:Box-iLQR

Aircraft Controls Researcher

Indian Institute of Science

May 2023 – July 2023 Bengaluru, India

- Designed **control scheduling** strategies for managing **nonlinear aircraft dynamics**, improving system stability across different flight phases.
- Identified and analyzed **equilibrium points** for multi-surface control aircraft models, successfully implementing linear control techniques to stabilize highly nonlinear flight conditions.
- Implemented the control algorithm and scheduling task in **Simulink**.

EDUCATION

PhD in Aerospace Engineering

Texas A&M University (4.0/4.0)

August 2023 – Ongoing

M.Tech. in Aerospace Engineering

Indian Institute of Technology, Kanpur (10.0/10.0)

January 2021 – December 2022

B.Tech. in Aerospace Engineering

Indian Institute of Technology, Kanpur (8.2/10.0)

July 2017 – May 2021

Relevant Coursework: Optimal Control

- Reinforcement Learning
- Linear Algebra
- Differential Equations

PUBLICATIONS/REPORTS

- "PPO and TD3 for Safe Reinforcement Learning Using Control Barrier Functions," Project Report, Reinforcement Learning, 2025.
- "Convexity in Optimal Control Problems," IEEE Conference on Decision and Control, 2024, Milan, Italy.
- "An Optimal Solution to Infinite Horizon Non-holonomic and Discounted Nonlinear Control Problems" American Control Conference 2025.
- "Optimal Control with Lyapunov Stability for Space Applications," AAS/AIAA Astrodynamics Specialist Conference, 2024, Colorado, USA.
- "Control Strategy for Trading Off Solar Power and Control Input," Journal of Aerospace Engineering, 2024.
- "Design and Path Optimization of a Spacecraft for Space Debris Removal ", 73rd International Astronautical Congress, Paris, France.

SKILLS

- Programming & Tools: MATLAB Simulink
- C C++ Python Labview CasADi
- PyTorch Git MuJoCo
- Robotics & Control: ROS iLQR DDP
- SQP MPC RL

HONORS AND AWARDS

- Aero Graduate Excellence Fellowship (2024,2025)
- Best Outgoing Sportsperson (IITK, 2023)
- Institute Silver Medal (IITK, 2019)
- Member of Tau Beta Pi (TAMU, 2024)